SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO5	**Assessment:**	Assessment Tool 1 – **Research Paper Review / Literature Survey**
Weightage:	20%	**Total Marks:**	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	G. Akshya	**Reg. No.:**	192425132
Date:	27/08/2026	**Duration:**	60 minutes

Code of Conduct

I G. Akshya (192425132) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *G. Akshya*

CO 5 Assessment Tool -02

Name: -G. Akshya **Reg No:** -192425132

Course Code: -CSA1007 **Slot:** -D

1. Introduction

Vertical farming enables crops to be cultivated in stacked layers within controlled environments, making it suitable for locations where land and water are limited. The proposed Smart Vertical Farming Automation Platform focuses on combining environmental monitoring, irrigation optimization, automated lighting, crop-health prediction, production reporting, and notifications in one integrated software system. Recent research shows that IoT-enabled vertical farming can improve monitoring and automation, while AI and machine learning can support prediction and decision-making.

2. Assigned Problem Statement

Develop a Smart Vertical Farming Automation Platform that continuously monitors environmental conditions, optimizes irrigation, predicts crop health, automates lighting control, generates production reports, and notifies farm operators. The intended outcome is higher crop yield, lower water consumption, improved resource efficiency, better crop quality, and sustainable food production.

3. Objectives

- Monitor environmental conditions such as temperature, humidity, light intensity, pH, nutrient concentration, and water-related parameters.
- Optimize irrigation and fertigation according to crop requirements and sensor readings.
- Predict crop health and support early identification of crop stress or nutrient-related problems.
- Automate lighting control according to crop growth requirements and environmental conditions.
- Generate production reports containing environmental readings, irrigation activity, crop-growth information, and resource usage.
- Notify farm operators when abnormal conditions, low water levels, sensor problems, or other important events are detected.

4. Literature Survey Method

The literature survey was organized around the requirements of the assigned problem statement. Research was reviewed for IoT-based environmental monitoring, automated irrigation and fertigation, lighting optimization, AI/ML-enabled vertical farming, and scalable smart-farm architectures. The assessment format requires individual paper reviews, comparative analysis, research gaps, future directions, and properly formatted references.

5. Individual Research Paper Reviews

5.1 Comparative analysis of IoT-based controlled environment and uncontrolled environment plant growth monitoring system for hydroponic indoor vertical farm

- **Research Problem:** Evaluate whether an IoT-enabled controlled vertical hydroponic environment improves plant growth compared with an uncontrolled/manual setup.
- **Methodology:** Developed an IoT-enabled vertical hydroponic system using Deep Flow Technique, sensors, and automated pH and TDS balancing; compared controlled and uncontrolled setups using Romaine lettuce.
- **Datasets / Tools / Technologies:** IoT sensors, hydroponic DFT setup, automated pH/TDS balancing.
- **Key Findings:** The automated setup produced higher observed fresh aerial plant weight (58.66 g) than the unautomated setup (48.81 g).
- **Limitations:** The study is focused on a specific hydroponic setup and crop, so broader generalization to different crops and larger multi-tier farms requires further validation.

- **Research Gap:** A broader platform can combine environmental monitoring with prediction, automated lighting, notifications, and production reporting instead of focusing mainly on environmental control.

5.2 Empowering vertical farming through IoT and AI-Driven technologies: A comprehensive review

- **Research Problem:** Review how IoT, AI, ML, and deep learning are being used in vertical farming.
- **Methodology:** Comprehensive review covering disease detection, crop-yield prediction, nutrition, irrigation control, sensing, cloud processing, and actuator-based automation.
- **Datasets / Tools / Technologies:** IoT, machine learning, deep learning, computer vision, cloud platforms, sensors and actuators.
- **Key Findings:** The review describes an automation pipeline in which sensors collect farm data, cloud/ML systems process it, and actuators can control water pumps, artificial lighting, and nutrient pumps.
- **Limitations:** The review identifies technological benefits but also indicates that integration, data quality, cost, and practical deployment remain important challenges.
- **Research Gap:** There is a need for a unified software architecture that combines monitoring, prediction, control, reporting, and operator alerts in a practical workflow.

5.3 Design, Development and Testing of an IoT-Based Smart Vertical Hydroponic System for Optimized Nutrient Management in a Controlled Environment

- **Research Problem:** Improve irrigation and nutrient management in a smart vertical hydroponic environment.
- **Methodology:** Developed and tested an IoT-based vertical NFT hydroponic system for basil using different nutrient-solution flow rates and fertigation management approaches.
- **Datasets / Tools / Technologies:** IoT-based fertigation, vertical NFT hydroponics, remote monitoring, nutrient-flow control.
- **Key Findings:** The study reported that a 1 lpm flow rate was suitable for the developed system. Compared with manual fertigation, the IoT-based system increased plant height, leaf area, number of leaves, fresh weight, yield, and water-use efficiency.
- **Limitations:** The experimental evaluation focuses on basil and specific flow/fertigation settings; lighting, crop-health prediction, and wider farm-level orchestration are not the central focus.
- **Research Gap:** An integrated platform can use sensor data and crop-health analytics to dynamically coordinate irrigation, lighting, alerts, and reporting.

5.4 Dynamic light optimization in vertical farming using an IoT-driven digital twin framework and artificial intelligence

- **Research Problem:** Optimize lighting recipes in vertical farming while improving automation and environmental control.
- **Methodology:** Developed a digital twin for a vertical-farming growth tower, connected it to an open-source IoT platform, and used a genetic algorithm to identify optimized lighting recipes for lettuce.
- **Datasets / Tools / Technologies:** Digital twin, IoT platform, artificial intelligence, genetic algorithm, LED lighting.
- **Key Findings:** The work demonstrates the use of digital-twin and AI techniques for automated lighting decision-making in vertical farming.
- **Limitations:** The main focus is lighting optimization, so other farm functions such as irrigation, crop-health prediction, and operator reporting are not treated as one integrated platform.
- **Research Gap:** Lighting optimization should be coordinated with water, environmental conditions, crop stage, energy consumption, and crop-health predictions.

5.5 Application of Artificial Intelligence and Machine Learning in Vertical Farming: A Comprehensive Review

- **Research Problem:** Systematically review the application of AI and ML in vertical farming and identify the transition toward data-driven farming.
- **Methodology:** Systematic review of 208 peer-reviewed studies published from 2015 to 2025, covering AI/ML applications in vertical farming.
- **Datasets / Tools / Technologies:** AI, machine learning, IoT, computer vision, predictive analytics and smart control approaches.
- **Key Findings:** The review describes the movement from rigid rule-based systems toward more flexible, data-driven operations that can adapt in real time.
- **Limitations:** A review does not itself provide a single implemented end-to-end platform, and real-world deployment still depends on reliable sensing, data quality, computing, energy, and integration.
- **Research Gap:** A practical student-scale platform can bridge the gap by implementing a complete sensing-to-decision-to-action workflow with explainable reports and alerts.

6. Comparative Literature Survey

Author / Year	Research Problem	Methodology	Tools / Dataset	Key Findings	Limitations	Research Gap
2023 study	IoT controlled vs uncontrolled vertical hydroponics	DFT + automated pH/TDS + comparison	IoT sensors; Romaine lettuce	Automated setup showed higher fresh weight	Specific crop/setup	Need integrated prediction + control + reporting
2024 review	IoT and AI applications in vertical farming	Review of sensing, ML, DL, cloud and actuation	IoT, ML, DL, CV	Supports automated irrigation, lighting and nutrition decisions	Integration/deployment challenges	Need unified end-to-end platform
Bhat et al., 2025	Optimize fertigation in vertical hydroponics	IoT-based NFT + flow-rate experiments	IoT fertigation; basil	Improved growth, yield and water-use efficiency	Crop-specific experiment	Need multi-function farm orchestration
2025 study	Optimize vertical-farm lighting	Digital twin + IoT + genetic algorithm	Digital twin; LED; lettuce	AI can optimize lighting recipes	Lighting-focused	Coordinate lighting with water, health and energy
Kim et al., 2026	Review AI/ML in vertical farming	Systematic review of 208 studies	AI, ML, IoT, CV	Shift toward adaptive data-driven farming	Review, not an end-to-end system	Practical integrated implementation needed

7. Critical Analysis

- IoT-based sensing is consistently identified as an important foundation for smart vertical farming because it enables continuous observation of environmental and nutrient-related conditions.
- Automated irrigation and fertigation can improve resource efficiency by applying water and nutrients according to measured requirements rather than relying entirely on manual operation.

- AI and machine learning provide an opportunity to move beyond simple threshold-based control toward predictive crop-health and decision-support functions.
- Lighting is an important control variable in indoor vertical farms, and recent work shows that AI and digital-twin approaches can be used to optimize lighting strategies.
- A major engineering challenge is integration: monitoring, irrigation, lighting, prediction, reporting, alerts, and energy/resource management are often investigated as separate components.
- Scalability and reliability are also important. A practical platform should handle multiple sensors and growing tiers while detecting sensor or actuator failures and notifying the operator.

8. Identified Research Gap

The literature indicates strong progress in individual smart-farming functions, but an important gap remains in building a unified vertical-farming automation platform that connects real-time environmental sensing, irrigation optimization, crop-health prediction, adaptive lighting, reporting, and operator notification within one software workflow. Existing studies also show the importance of resource efficiency, energy management, reliable sensing, and scalable architectures. Therefore, the proposed platform can focus on an integrated, modular architecture that supports data collection, analysis, automated action, and human oversight.

10. Proposed System Approach

- Sensor Layer – collect temperature, humidity, light intensity, pH, TDS/EC, water level, and other selected parameters.
- Data Acquisition Layer – transmit sensor readings through a microcontroller such as ESP32 to a local or cloud database.
- Monitoring Dashboard – display current environmental conditions, historical trends, alerts, and farm status.
- Decision and Prediction Layer – apply threshold/rule-based logic initially and machine-learning models where sufficient historical data are available.
- Automation Layer – control water pumps, nutrient delivery, and LED lighting based on crop requirements and system decisions.
- Alert Layer – send notifications when readings cross safe limits, resources are low, or abnormal system conditions are detected.
- Reporting Layer – generate daily/weekly production and resource-use reports to support farm management decisions.

11. Future Research Directions

- Develop crop-specific machine-learning models for early crop-stress and health prediction.
- Use computer vision to analyze plant images for growth and disease/stress indicators.
- Integrate energy-aware lighting optimization to reduce the electricity cost of indoor farming.
- Add fault detection for sensor, pump, lighting, and communication failures.
- Use predictive irrigation instead of only threshold-based irrigation.
- Support multiple vertical layers and multiple crop types through a scalable software architecture.
- Integrate renewable energy sources and energy-management strategies for more sustainable operation.
- Provide explainable recommendations so operators can understand why irrigation, lighting, or alerts were triggered.

11. Conclusion

The literature survey shows that IoT, automation, AI, machine learning, digital twins, and smart sensing can significantly strengthen vertical farming systems. Existing research demonstrates benefits in environmental monitoring, automated fertigation, lighting optimization, and data-driven decision-making. However, the reviewed approaches also reveal a need for integrated systems that combine these functions with crop-health prediction, reporting, alerts, and resource-aware control. The proposed Smart Vertical Farming Automation Platform addresses this gap by providing a modular end-to-end approach intended to improve crop yield, reduce water consumption, increase resource efficiency, improve crop quality, and support sustainable food production.

12. References

- Environmental Research (2023). Comparative analysis of IoT-based controlled environment and uncontrolled environment plant growth monitoring system for hydroponic indoor vertical farm. Volume 222, Article 115313. DOI: 10.1016/j.envres.2023.115313.
- Heliyon (2024). Empowering vertical farming through IoT and AI-Driven technologies: A comprehensive review. DOI: 10.1016/j.heliyon.2024.e34998.
- Bhat, A. G., Hasan, M., Singh, D. K., Sahoo, R. N., et al. (2025). Design, Development and Testing of an IoT-Based Smart Vertical Hydroponic System for Optimized Nutrient Management in a Controlled Environment. Irrigation and Drainage, 74(5), 1904–1920. DOI: 10.1002/ird.4002.
- Applied Soft Computing (2025). Dynamic light optimization in vertical farming using an IoT-driven digital twin framework and artificial intelligence. Volume 174, Article 112985. DOI: 10.1016/j.asoc.2025.112985.
- Kim, M. Y., Park, G., & Seo, C. H. (2026). Application of Artificial Intelligence and Machine Learning in Vertical Farming: A Comprehensive Review. Sustainability, 18(16), 8261. DOI: 10.3390/su18168261.

- Wei En, L., Lim, C. L., Kok, C. L., Koh, Y. Y., Lee, C. C. C., et al. (2026). Sustainable urban farming using a smart hydroponic approach using IoT and real time monitoring. Scientific Reports.
- Autonomous smart vertical farming: A systems-level synthesis of IoT and solar energy integration for sustainable agriculture (2026). Energy Reports. DOI: 10.1016/j.egyr.2026.109509.